

Coral biomimetic PDMS marine antifouling coating based on the intrinsic antibacterial and fluorescent response synergistic strategies

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ABSTRACT

The applicability of polydimethylsiloxane (PDMS)-based marine antifouling coatings is restricted by their limited mechanical properties and static antifouling capabilities. In this study, we have developed a coral biomimetic PDMS marine antifouling coating, designated as PDMS-DAF-x-AMC, which employs synergistic antifouling strategies based on intrinsic antibacterial and fluorescent response mechanisms. Here, 'x' represents the amount of 3,4-diaminofurazan (DAF) incorporated into the formulation. Hydroxyl-terminated polydimethylsiloxane was employed as the primary matrix with DAF serving as chain extenders, 7-amino-4-methylcoumarin (AMC) acting as the end-capping agent and zinc ions as the coordinating metal. The dosages of DAF are 0.0, 0.1, 0.2, and 0.3 g, respectively, while the dosage of AMC is consistently maintained at 0.03 g. The supramolecular interactions, including hydrogen bonds and coordination bonds within the PDMS network enhance the mechanical properties, adhesion strength, and self-healing capabilities. Furthermore, the inherent antibacterial activity of DAF, combined with the fluorescence response of AMC synergistically imparts the coating with superior fouling-resistant properties with a bacteriostatic efficiency exceeding 99 % and maintaining a minimum diatom density of 18 ± 11 diatoms mm^{-2} . Also, the low surface energy ranging from 21.11 to 24.86 $\text{mJ}\cdot\text{m}^{-2}$ gave the coating excellent self-cleaning properties and promoted the fouling-release performance. This study aims to introduce an innovative design concept for the advancement of high-performance PDMS antifouling coatings through the incorporation of various environmentally sustainable biomimetic antifouling strategies.

1. Introduction

Marine biofouling is a phenomenon characterized by the adhesion of marine organisms to surfaces [1,2]. The adherence of fouling organisms to vessel surfaces can diminish navigational speed, consequently elevating fuel consumption. Additionally, biofouling can cause bio-corrosion of metallic structures and pose a risk of biological invasion [3–5]. Antifouling coatings have developed rapidly for their efficiency, cost-effectiveness, and ease of maintenance in mitigating biofouling issues related to marine operations. [6–8]. Nevertheless, conventional antifouling coatings typically incorporate toxic metals or biocides, which have been shown to negatively impact the marine ecosystem [9,10]. Therefore, the development of eco-friendly antifouling coatings is of paramount importance in safeguarding marine ecosystems [11].

Polydimethylsiloxane (PDMS) materials have increasingly been

recognized as an optimal choice for fouling release coatings (FRCs) due to their low surface energy, low elastic modulus, and low surface roughness. These characteristics facilitate the removal of fouling organisms through the shear forces generated by water flow [12,13]. However, the intrinsic mechanical fragility of PDMS-based coatings makes them prone to damage, thereby diminishing their overall durability [14]. The intrinsically low surface energy results in suboptimal adhesion to substrates, making them prone to detachment following prolonged immersion in seawater [15]. Moreover, the insufficient antifouling performance under static is ineffective in preventing the proliferation of bacteria and algae on their surfaces [16]. Consequently, augmenting the static antifouling properties and enhancing long-term durability in complex and potentially extreme marine environments present significant challenges that require extensive research and innovative solutions.

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To achieve optimal adhesion to the substrate while preserving low surface energy characteristics, a variety of soft segments with differing polarities were utilized. The integration of urethane/urea functional groups into the PDMS backbone has been observed to improve mechanical and adhesive properties by promoting hydrogen bond formation either between polymer chain segments or with the substrate [17–19]. Furthermore, drawing inspiration from the self-healing properties observed in living organisms, the incorporation of self-healing mechanisms into antifouling coatings represents a promising strategy for improving long-term durability [20,21].

The incorporation of non-ionic hydrophilic groups, amphiphilic compounds, or zwitterions into PDMS chains could enhance the static antifouling properties [22–24]. This modification facilitates the formation of a hydration layer on the coating surface, thereby significantly inhibiting the adsorption of non-specific proteins, reducing bacterial adhesion, and preventing biofilm formation. In addition, the reactive antifouling groups with biological activities can be introduced into the flexible polymer main chains by chemical grafting, thereby significantly enhancing the fouling resistance of coating surface [25–27]. In our previous study, we engineered a self-healing polydimethylsiloxane (PDMS)-based antifouling coating. The incorporation of flexible covalent disulfide bonds alongside strong non-covalent crosslinked multiple hydrogen bonds endows the coating with exceptional mechanical and self-healing characteristics, achieving adhesion strengths to steel substrates of up to 3.33 MPa. Importantly, marine field testing has revealed that the chemical grafting of benzothiazole antimicrobial groups onto the PDMS chain segments significantly enhances the static antifouling performance [28].

Numerous animal and plant species have evolved surfaces with remarkable antifouling abilities. Inspired by these biological features, including microstructures, wettability, and secretion of lubricants or natural antifoulants, researchers have created dynamic antifouling surfaces using biomimetic strategies [29]. Corals, categorized as sessile organisms, exhibit an absence of mobility in reaction to external stimuli; however, they maintain consistently sustain active and clean surfaces. Antifouling coatings inspired by these biological strategies could offer protective advantages for anchored vessels and stationary structures in aquatic environments [30]. Previous studies have demonstrated that corals have evolved five primary antifouling strategies: the production of antifoulants [31], the development of fouling-release surfaces [32], the secretion of sloughing slime [33], the utilization of soft tentacles [34], and the employment of fluorescent effects [35]. The mechanism of fluorescent antifouling was based on the relationship between cell physiological activities and light [36]. The fluorescence effect may impede diatom adhesion, whereas low light levels can supply the necessary energy for diatom photosynthesis, thereby affecting their settlement [37]. Jin et al. [35] developed an innovative PDMS-phosphor-silicone rubber sandwich structure coating characterized by its ability to emit low-intensity light during nighttime and its hydrophilic surface properties. The findings indicated that the surface hydrophilicity significantly diminished the adhesion of algae and bacteria. Furthermore, the study revealed that fluorescence selectively inhibited algal growth, with the blue (427 nm), yellow (663 nm), and red (731 nm) phosphor layers proving effective in mitigating the proliferation of *Chlorella* and *N. closterium f. minutissima*. Guo et al. [38] synthesized a polyurethane incorporating PDMS and polytetrahydrofuran as primary polymer chains, with 7-amino-4-methylcoumarin serving as an antifouling moiety through condensation polymerization. The characteristic benzopyrone structure of 7-amino-4-methylcoumarin not only facilitates a fluorescence response to ultraviolet radiation in seawater with high transmittance but also confers antibacterial properties [39,40]. Owing to the significant synergistic interaction between fluorescence antifouling and contact bacteriostasis mechanisms, the PDMS-based fouling-release surface exhibits exceptional antifouling performance with a minimal incorporation of the antifouling group (<0.20 wt%), representing one of the lowest concentrations reported in previous

studies.

Herein, we explored a novel coral biomimetic PDMS marine antifouling coating based on the intrinsic antibacterial and fluorescent response synergistic strategies. The coating was fabricated using hydroxyl-terminated polydimethylsiloxane (OH-PDMS) as the primary matrix, with 3,4-diaminofurazan (DAF) serving as chain extenders, and 7-amino-4-methylcoumarin (AMC) as the capping agent. Zinc ions (Zn^{2+}) were introduced to form metal-coordination bonds with DAF, while hydrogen-bonding interactions among the generated amine ester groups facilitate cross-linking and curing. The supramolecular interactions of hydrogen bonds and metal-coordination bonds endow the coating with enhanced mechanical properties, adhesion strength, and self-healing capabilities. Additionally, DAF is a compound distinguished by its 1,2,5-oxadiazole structure. This five-membered heterocyclic framework, which includes two nitrogen atoms and one oxygen atom, endows 1,2,5-oxadiazole and its derivatives with efficacy as pharmacophores, particularly in their application as carbonic anhydrase inhibitors [41], antibacterial [42], anti-protozoal [43], herbicides [44], and anticancer [45], among others [46]. However, the potential of 1,2,5-oxadiazole in the development of flexible antibacterial materials has received limited attention. As a result, the inherent antibacterial characteristics of the 1,2,5-oxadiazole structure within PDMS, combined with the fluorescent antifouling capabilities of AMC, work synergistically to inhibit the attachment of fouling organisms. It is anticipated that this biomimetic strategy, grounded in the principle of synergistic antifouling, will offer new perspectives for the advancement of environmentally friendly, high-performance PDMS-based antifouling coatings.

2. Experimental section

2.1. Reagents and materials

Hydroxyl-terminated polydimethylsiloxane (OH-PDMS, $M_n = 2000 \text{ g} \cdot \text{mol}^{-1}$) was purchased from Jiande polymer new materials Co., Ltd. and subjected to rotary evaporation at 80°C for 2 h prior to utilization. Isophorone diisocyanate (IPDI, 99 %), dibutyltin dilaurate (DBTDL, 95 %), 1,4-butanediol (BDO, 98 %), 3,4-diaminofurazan (DAF, 98 %), 7-amino-4-methylcoumarin (AMC, 98 %), tetrahydrofuran (THF, $\geq 99.5\%$) and ZnCl_2 (98 %) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. Sylgard 184 PDMS was purchased from Dow Corning. Artificial seawater (ASW) was prepared according to ASTM D1141. All other reagents were used as received from the manufacturer unless otherwise noted.

2.2. Synthesis of prepolymers

Taking PDMS-DAF-0.3-AMC as a representative case, a mixture of OH-PDMS (10 g, 5 mmol), BDO (0.45 g, 5 mmol), and IPDI (3.16 g, 14.2 mmol) was dissolved in THF and introduced into a 250 mL three-necked flask equipped with a mechanical stirring apparatus, a nitrogen circulation system, and a condensation setup. The reaction was conducted at 65°C for 2 h to synthesize an isocyanate-terminated prepolymer. Subsequently, the solution was allowed to cool to ambient temperature before the addition of DAF (0.3 g, 3 mmol), and the reaction was continued for an additional 2 h. The temperature was elevated to 70°C , followed by the addition of AMC (0.03 g, 0.17 mmol), and the reaction was allowed to proceed for 3 h. ZnCl_2 (0.136 g, 1 mmol) was dissolved in THF and introduced into the reaction system, where it was stirred for 1 h to facilitate the formation of a metal-coordination bond. The resulting prepolymers were designated as PDMS-DAF-x-AMC, where x denotes the molar amount (in mmol) of 3,4-diaminofurazan (DAF) added. The molar ratio of metal to ligand (Zn^{2+} : DAF) was consistently maintained at 1:3.

2.3. Preparation of PDMS-DAF-x-AMC coatings

The glass slides and tinplate substrates underwent ultrasonic

cleaning using acetone and ethanol for a duration of 10 min, followed by rinsing with deionized water. Subsequently, PDMS-DAF-x-AMC was dissolved in THF to formulate a polymer solution with a solid content of 30 %, the polymer solution was subsequently ultrasonically agitated for 30 min to obtain a uniform dispersion to ensure the uniform distribution of zinc ions in the solution. This solution was then applied as a coating onto the glass slides and tinplate. The coated substrates were placed in a drying oven for 48 h to facilitate complete solvent evaporation. Tensile tests were conducted utilizing PDMS-DAF-x-AMC samples applied to polytetrafluoroethylene molds. Coatings on glass slides and tinplate were employed for various assessments, including water contact angle measurement, pseudobarnacle removal tests, adhesion tests (with a thickness range of 200–300 μm), antibacterial and anti-diatom evaluation.

2.4. Characterization

Fourier transform infrared spectroscopy (FTIR) analyses were conducted on a Bruker TENSOR-27 infrared spectrophotometer using the KBr disc method. Each spectrum was based on 32 scans between 4000 and 500 cm^{-1} . ^1H NMR was recorded on a Bruker Avance 500 MHz NMR spectrometer at 25 $^{\circ}\text{C}$ using deuterated chloroform (CDCl_3) as the solvent. The Raman spectra were obtained utilizing a Horiba Scientific LabRAM HR Evolution spectrometer, employing an excitation wavelength of 785 nm, with a wavenumber range of 200 to 4000 cm^{-1} . The transmittance of the coating was assessed using an ultraviolet spectrophotometer manufactured by Shanghai Spectral Element Instrument Co., Ltd., with a testing range of 300 to 800 nm. The fluorescence emission spectra were obtained by exciting the samples at their respective absorption maxima, while the excitation spectra were acquired by monitoring the fluorescence emission maxima. A UV lamp with a wavelength of 365 nm and a power output of 6 watts was utilized as the source of UV irradiation. The energy dispersive X-ray spectrum (EDS) element diagram was studied by using JSM-6700F scanning electron microscope of JEOL company in Japan.

The tensile properties and self-healing tests were conducted using a microcomputer-controlled electronic universal testing machine, with the testing rate maintained at 50 mm/min and the temperature set to ambient conditions. During the cyclic tensile testing, samples were unloaded immediately upon reaching 300 % strain. Each sample underwent ten consecutive stretching cycles without any interval between cycles. The self-healing efficiency was determined using Eq. (1).

$$\eta = \frac{\sigma_{\text{healed}}}{\sigma_{\text{pristine}}} \times 100\% \quad (1)$$

Here, σ_{pristine} and σ_{healed} correspond to the tensile strength of the original sample and the self-healing one, respectively. The shore hardness was evaluated using an LXA hardness tester (American ABE) under ambient temperature conditions. Dynamic mechanical analysis (DMA) was carried out on a Q800 (TA Instrument) with film tension clamps from -125 to 40 $^{\circ}\text{C}$ at a heating rate of 3 $^{\circ}\text{C min}^{-1}$. Thermogravimetric analysis (TGA) was conducted using a SDTQ600 thermal analyzer, employing a heating rate of 10 $^{\circ}\text{C min}^{-1}$ under a nitrogen atmosphere. The analysis was performed from ambient temperature to 600 $^{\circ}\text{C}$, with a nitrogen flow rate of 50 mL min^{-1} and a sample mass of approximately 10 mg.

The static water contact angles and surface free energy were determined by a contact angle goniometer (OCA 25, Dataphysics, Germany) by depositing 3 μL water or diiodomethane droplets on the coating using the sessile method at 25 $^{\circ}\text{C}$ and 52 % relative humidity, five measurements were performed on different regions on each sample and averaged. The surface free energy was calculated using the Owens-Wendt-Rabel-Kaelble method.

To quantify the cumulative release of Zn^{2+} from the PDMS-DAF-0.3-AMC coating surface, the prepared coating was immersed in artificial seawater (ASW) for 1, 3, and 7 days. Subsequently, 10 mL of the

immersion solution was collected at each time point as a test sample. The concentration of Zn^{2+} in these samples was determined using graphite furnace atomic absorption spectrometry (GFAAS) on a TAS-990 atomic absorption spectrophotometer (Beijing Purkinje General Instrument Co., Ltd., China).

The adhesion strength on both tinplate and glass substrates was evaluated using a universal testing machine. The tensile rate was maintained at a diameter of 10 mm/min, with a designated measuring area of 20 mm. Each specimen underwent testing five times under consistent conditions, and the mean value of these trials was calculated for analysis. The pseudobarnacle removal strength tests were conducted in accordance with the ASTM D5618-20 (2020) standard. To execute these tests, five cylindrical aluminum studs with a diameter and height of 10 mm each were affixed to various regions of the coating surface. This attachment was achieved using a two-component epoxy adhesive (Araldite), and the adhesive was allowed to cure for a duration of 3 d under ambient conditions. Subsequently, a shear force was systematically applied to the aluminum studs in parallel to the surface, employing a force gauge (SUNDOO, SH-500), and maintained at a constant speed. Carbon powder and copper chloride were utilized as hydrophilic and hydrophobic pollutants, respectively, and were applied to the coating. Following an exposure period of 1 h, the pollutants were effectively removed using water droplets.

The antibacterial properties were evaluated utilizing the plate counting technique. Under sterile conditions, the samples were positioned in a 6-well plate, and subsequently, 5 mL of *Pseudomonas* and *Escherichia coli* (*E. coli*) suspensions, each with an optical density (OD) value of 1, were added. The samples were incubated at 37 $^{\circ}\text{C}$ for 4 h. Subsequently, the samples were transferred to centrifuge tubes containing 10 mL of phosphate-buffered saline (PBS) solution and subjected to ultrasonic treatment for 10 min. The culture solution was then diluted 1000-fold, and 50 μL of the diluted solution was evenly spread onto an agar solid medium. The plates were incubated at 37 $^{\circ}\text{C}$ for 12 h to allow for bacterial growth. Finally, remove the samples and capture photographic documentation. Obtain three parallel samples from each group and establish a blank control group (PDMS). Subsequently, determine the antibacterial efficacy by quantifying the colonies formed by *Pseudomonas* and *E. coli*. The calculation formula is as follows:

$$N = \frac{N_a - N_b}{N_a} \times 100\% \quad (2)$$

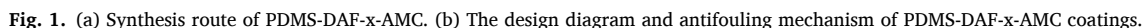
where N_a was the number of colonies in the control group and N_b was the number of colonies in the antibacterial group. Meanwhile, the samples with adhered *Pseudomonas* were fixed by glutaraldehyde (2.5 %) and washed with graded concentrations of ethanol, and SEM observed the morphology of dead *Pseudomonas*.

The marine diatom *Navicula incerta* (*N. incerta*) was obtained from the State Key Laboratory for Marine Corrosion and Protection in Qingdao. It was cultivated in an F/2 culture medium under a 12:12 h light-dark cycle at 23 $^{\circ}\text{C}$ in a light incubator. PDMS and PDMS-DAF-x-AMC were put into diatom suspension (1.2×10^6 cells mL^{-1}) at 25 $^{\circ}\text{C}$ with the illumination of natural light and using dark conditions as a control. After 7 d, the samples were gently rinsed with seawater to remove unsettled diatoms and then observed under a fluorescence detector (BX-51, Olympus, Japan) and the average value of the density of the attached diatoms was recorded.

Atomic Force Microscopy (AFM) images of coated cover glass slips were acquired by an atomic force microscope (Park Systems, Model XE-100) using a non-contact mode. The topography images were determined from a scan area of 5×5 μm^2 from three different regions of each sample. NanoScope Analysis 1.5 software was used to process and analyze the images. The value of R_q (the root mean square of the surface roughness height) was used to quantify the surface roughness of coatings.

The marine field test was conducted at the Qingdao Port (24 $^{\circ}$ 33'N,

Raman spectroscopy was employed to verify the formation of metal-coordination (Fig. 2(b)). The sample containing Zn^{2+} -coordination exhibited two peaks at 1152 cm^{-1} and 1511 cm^{-1} , which are indicative of coordination bonds formed by $\text{Zn}-\text{O}$ and $\text{Zn}-\text{N}$ interactions. The X-ray photoelectron spectroscopy (XPS) analysis was utilized to substantiate the specific interactions between Zn^{2+} and O/N elements. By distinguishing from the deconvoluted peaks in the XPS spectra of PDMS-DAF-0.3-AMC, a peak of $\text{Zn}^{2+} \bullet \bullet \bullet \text{N}$ (399.8 eV, Fig. S2a) in high resolution N 1 s spectrum and a peak of $\text{Zn}^{2+} \bullet \bullet \bullet \text{O}$ (531.9 eV, Fig. S2b) in high resolution O 1s spectrum were exposed [48,49]. The spectral results,



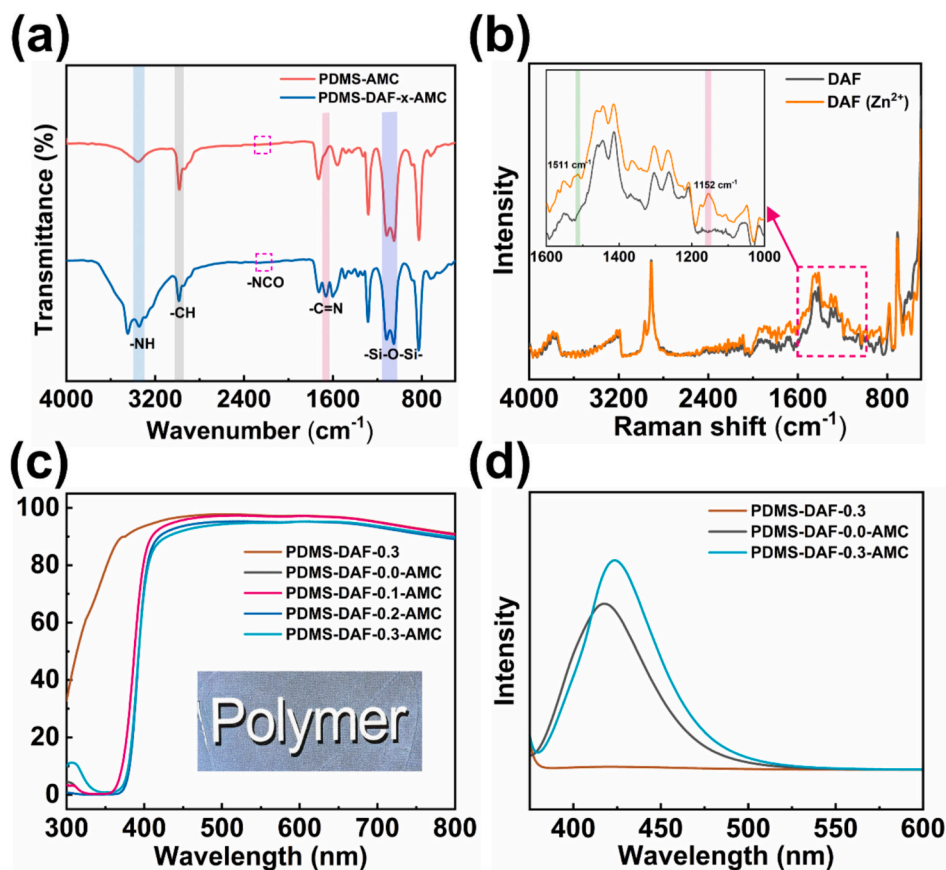


Fig. 2. (a) FTIR spectra of PDMS-DAF and PDMS-DAF-x-AMC. (b) Raman spectra of PDMS-DAF-x-AMC, with the red spectrum indicating Zn²⁺ coordination and the black spectrum indicating the absence of Zn²⁺ coordination. (c) The transparency of PDMS-DAF-0.3 and PDMS-DAF-x-AMC is presented, along with a photograph of the PDMS-DAF-0.3-AMC coating applied to a PET substrate. (d) Fluorescence emission spectra of the coatings with and without the incorporation of AMC (at the excitation wavelength of 365 nm). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

corroborated by Raman spectrum analysis, provide compelling evidence for the presence of Zn (II) coordination in PDMS-DAF-0.3-AMC, thereby substantiating the dual coordination structure of Zn²⁺ with DAF on the coating surface.

As illustrated in Fig. 2(c), PDMS-DAF-x-AMC exhibited significant transparency across the visible spectrum range (470–800 nm). Owing to the fluorescence response of AMC, the sample showing maximum absorption at 365 nm. Additionally, the fluorescence emission spectra of the coatings with and without the incorporation of AMC are presented in Fig. 2(d). A UV lamp with a wavelength of 365 nm and a power output of 6 watts was employed as the source of UV irradiation. It is evident that PDMS-DAF-0.0-AMC and PDMS-DAF-0.3-AMC exhibit characteristic emissions at 418 nm and 424 nm, respectively. Nonetheless, the fluorescence intensity of PDMS-DAF-0.3 remained unchanged, thereby further substantiating the fluorescence response of the coating upon the introduction of AMC. The EDS elemental mapping further reveals a uniform distribution of carbon (C), nitrogen (N), oxygen (O), silicon (Si), and zinc (Zn) across the surface (Fig. S3). The uniform distribution of zinc ions further corroborates the successful formation of metal coordination. Additionally, the EDS point scan provides a clear depiction of the presence and relative abundance of C, N, O, Si, and Zn.

3.2. Mechanical and self-healing properties

Fig. 3(a) presents the representative tensile stress-strain curves alongside the corresponding elongated images of PDMS-DAF-0.3-AMC at different stages of deformation. Notably, the elongation at break for PDMS-DAF-0.0-AMC exceeds 1000 %. Despite this significant elongation, the coating demonstrates a relatively low stress response, with a

tensile strength of only 0.3 MPa. However, the coating demonstrates low stress and a tensile strength of only 0.3 MPa. This phenomenon can be attributed to the reduced content of hard segments and the limited presence of hydrogen bonds. In contrast, an increase in DAF content results in a marked enhancement of tensile strength. Furthermore, the coordination effect of Zn²⁺ contributes to the improvement of mechanical properties. The tensile strength of PDMS-DAF-0.3-AMC was determined to be 2.78 MPa, while the elongation at break was recorded at 590 %. As illustrated in Fig. 3(b), there is a positive correlation between the Young's modulus of PDMS-DAF-x-AMC and the DAF content. The toughness of the PDMS-DAF-x-AMC was evaluated by integrating the stress-strain curve with the tensile toughness represented by the area under the curve. The findings indicate a monotonic increase in toughness over time. Therefore, it is hypothesized that the increased DAF content, along with the presence of metal coordination bonds, exerts a beneficial effect on the mechanical properties of PDMS-DAF-x-AMC. The integration of carbamate bonds and urea groups as hard segments is anticipated to improve the elastomer's hardness. As illustrated in Fig. S4, a progressive increase in hardness is observed.

The influence of DAF on the dynamic mechanical properties of PDMS-DAF-x-AMC was examined by analyzing the correlation between storage modulus (E') and temperature. As illustrated in Fig. 3(c), PDMS-DAF-x-AMC exhibits a glassy state characterized by a high E' value within the temperature range of -130 to -120 °C. With increasing temperature, E' decreases, suggesting that elevated temperatures enhance viscosity and fluidity. Furthermore, PDMS-DAF-x-AMC undergoes a transition from a glassy state to a rubbery state. Fig. 3(d) illustrates the peak value of the Tan δ response for PDMS-DAF-x-AMC to denote the glass transition temperature (T_g). An increase in the

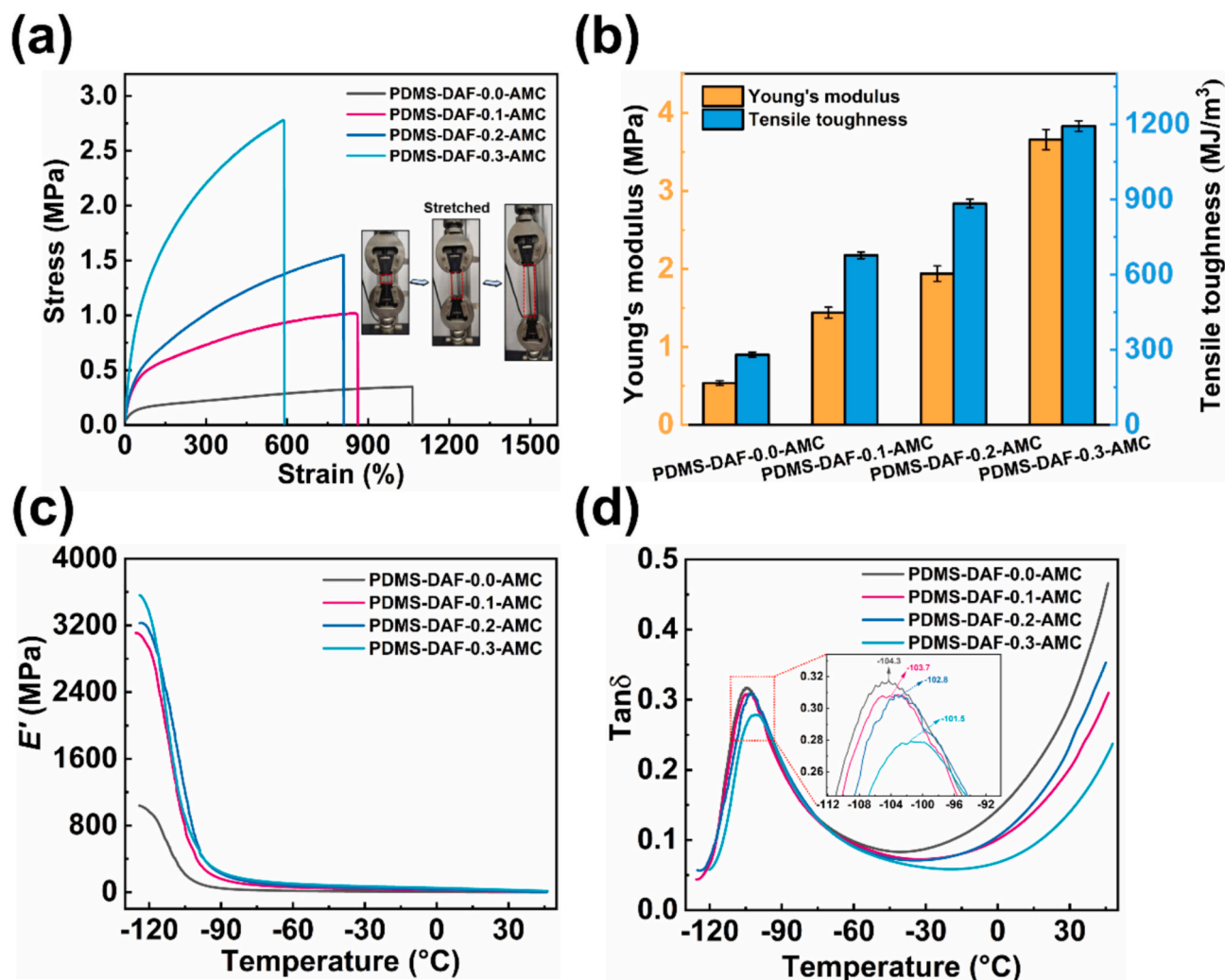


Fig. 3. (a) Stress-strain curves and optical pictures of PDMS-DAF-x-AMC stretching process. (b) Young's modulus and tensile toughness of PDMS-DAF-x-AMC. (c) E' -temperature curves, and (d) $\tan \delta$ -temperature curves PDMS-DAF-x-AMC.

concentration of DAF is associated with an augmented formation of hydrogen and coordination bonds, resulting in a slight elevation in T_g .

In order to examine the influence of DAF on the formation of hydrogen bonds, we performed simulations of the molecular structure of PDMS-DAF-x-AMC using the Materials Studio software. The Forcite module was employed to carry out structure optimization, dynamic annealing, and dynamic simulations under both NVT and NPT ensembles. The results indicated that the simulated PDMS-DAF-x-AMC achieved a highly stable structural configuration. Fig. 4(a) illustrates that the blue dotted line in the magnified local view denotes the hydrogen bond formed between the carbamate and urea groups. In Fig. 4(b), it is evident that the number of hydrogen bonds increases with the DAF content. It is well-documented that metal coordination can significantly enhance the mechanical properties of materials. Accordingly, we employed molecular simulation software to analyze the bond energy associated with metal coordination, as illustrated in Fig. 4(c), which presents the simulation diagram of Zn^{2+} coordination. Employing the energy calculation function within the Forcite module, as shown in Fig. 4(d), it was observed that following the formation of the coordination bond, the overall bond energy increased from 380.17 kcal/mol to 594.65 kcal/mol. This finding indicates that the coating displays improved stability of the cross-linked network following metal coordination.

In addition, cyclic tensile tests were conducted to assess the self-recoverability of PDMS-DAF-x-AMC (Fig. S5). The tensile testing

comprised five cycles, each subjected to a strain of 300 %. The energy dissipation for PDMS-DAF-0.0-AMC was determined to be 36.5 MJ m⁻³. Upon the incorporation of DAF and the implementation of metal ion coordination resulted in a substantial enhancement in energy dissipation, with observed values reaching 123 MJ m⁻³, 144 MJ m⁻³, and 345 MJ m⁻³. In each sample, significant hysteresis was observed during the initial loading-unloading cycle, indicating that energy dissipation occurred as a result of the fracture of hydrogen bonds and metal coordination bonds, as well as the rearrangement of molecular segments within the chains. The hysteresis loop observed in the second cycle was significantly reduced compared to the first cycle. This reduction is attributed to the fact that the rearranged molecular fragments and broken non-covalent bonds did not have sufficient time to recover within the short interval, leading to a progressive decrease in energy loss during subsequent stretching processes. The presence of hydrogen bonds and metal coordination bonds enhances rapid self-healing at 60 °C within 2 h. As illustrated in Fig. S6, both macroscopic and microscopic images depict the PDMS-DAF-0.3-AMC before and after the healing process. Post-healing, the PDMS-DAF-0.3-AMC exhibits a stress capacity of 2.34 MPa, achieving a self-healing efficiency of 85 %. The restored sample is capable of effortlessly lifting a weight of 500 g.

The progressive enhancement in tensile strength (0.3 to 2.78 MPa) and Young's modulus (0.38 to 4.21 MPa) with increasing DAF content is attributed to three interlinked structural factors: FTIR spectra (Fig. 2a) confirm the formation of urea groups (-NH-CO-NH-) through intensified

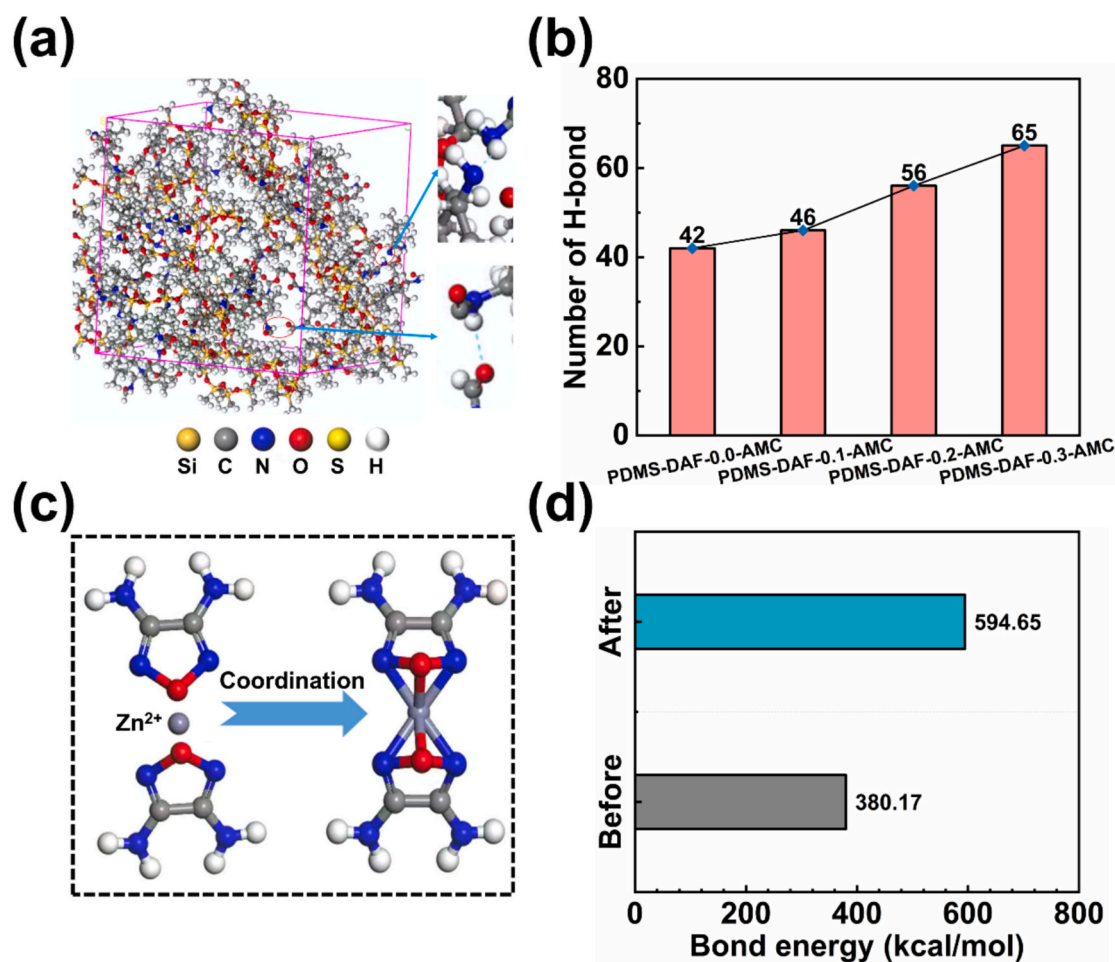


Fig. 4. (a) Amorphous cell of PDMS-DAF-x-AMC. (b) Number of hydrogen bonds in amorphous cells of PDMS-DAF-x-AMC. (c) Coordination between DAF and Zn²⁺. (d) Bond energy of coordination bond formed by DAF and Zn²⁺.

N—H (3350 cm⁻¹) and C=O (1660 cm⁻¹) vibrations, creating rigid hard segments that elevate the storage modulus by 37 % (Fig. 3c); Molecular simulations (Fig. 4b) quantify a 217 % increase in hydrogen bond density (12 to 38 bonds/nm³), directly correlating with improved energy dissipation during cyclic stretching (Fig. S5); Raman and XPS data (Figs. 2b, S2) verify Zn²⁺ coordination with DAF's N/O sites, forming high-energy bonds (594.65 kcal/mol, Fig. 4d) that stabilize the supra-molecular network, as evidenced by the Shore hardness increase from 25 A to 38 A (Fig. S4). These chemically anchored hard segments and dynamically reversible bonds collectively enable simultaneous strength and self-healing capabilities.

3.3. Surface properties and adhesion strength

The wettability of the coating surface was assessed utilizing an optical contact angle analyzer. As illustrated in Fig. 5(a), contact angle measurements with water were documented. Due to the intrinsic hydrophobic properties of PDMS, the PDMS-DAF-x-AMC coating demonstrated a hydrophobic surface, with the hydrophobicity progressively enhancing. Significantly, the maximum water contact angle recorded for PDMS-DAF-0.3-AMC was approximately 105.8°, reflecting a substantial degree of hydrophobicity.

The surface energy (SE) of the PDMS-DAF-x-AMC coating was determined using the Fowkes equation. As shown in Fig. 5(b), the SE ranged from 21.11 to 24.86 mJ m⁻². The study demonstrated that a SE below 30 mJ m⁻² effectively inhibits the adhesion of marine organisms, such as mussels and barnacles. Moreover, when the surface energy is

below 25 mJ m⁻², the coating exhibits excellent fouling-release properties [50]. Fig. S7 illustrates the cumulative release of Zn²⁺ ions from the PDMS-DAF-0.3-AMC coating over various time intervals. Following immersion in artificial seawater (ASW) for durations of 1 day and 3 days, the Zn²⁺ concentrations in the solution were measured at 0.803 mg L⁻¹ and 1.812 mg L⁻¹, respectively. The data indicate a linear increase in cumulative Zn²⁺ release with extended exposure time. The observed rapid release after 3 days is attributed to the facile dissolution of Zn²⁺ ions dispersed on the coating's surface into the aqueous environment. However, the Zn²⁺ concentration value of the solution was 1.824 mg L⁻¹ after the coating immersed in ASW for 7 days. The concentration of Zn²⁺ has shown minimal increase, indicating that no additional Zn²⁺ have been released from the coating surface. This suggests that the Zn²⁺ within the coating remain coordinated with DAF through coordination bonds, thereby maintaining the stability of the coating.

Ensuring robust adhesion between the coating and the substrate surface is crucial for maintaining the coating's integrity under various environmental conditions. The adhesion strength of PDMS-DAF-x-AMC on glass and tinplate substrates was assessed using pull-off testing, as illustrated in Fig. 5(c). The adhesion strength to the glass slide ranges from a minimum of 1.03 MPa to a maximum of 1.5 MPa. This variation is attributed to the hydrogen bond interactions between the PDMS-DAF-x-AMC coating and the substrate surface, which confer substantial adhesion strength. Furthermore, the adhesion of the coating is positively correlated with the DAF content. The adhesion strength of the coating on the tinplate sheet exceeded that on glass, exhibiting values between 1.36 MPa and 2.0 MPa. The improved adhesion is ascribed to the

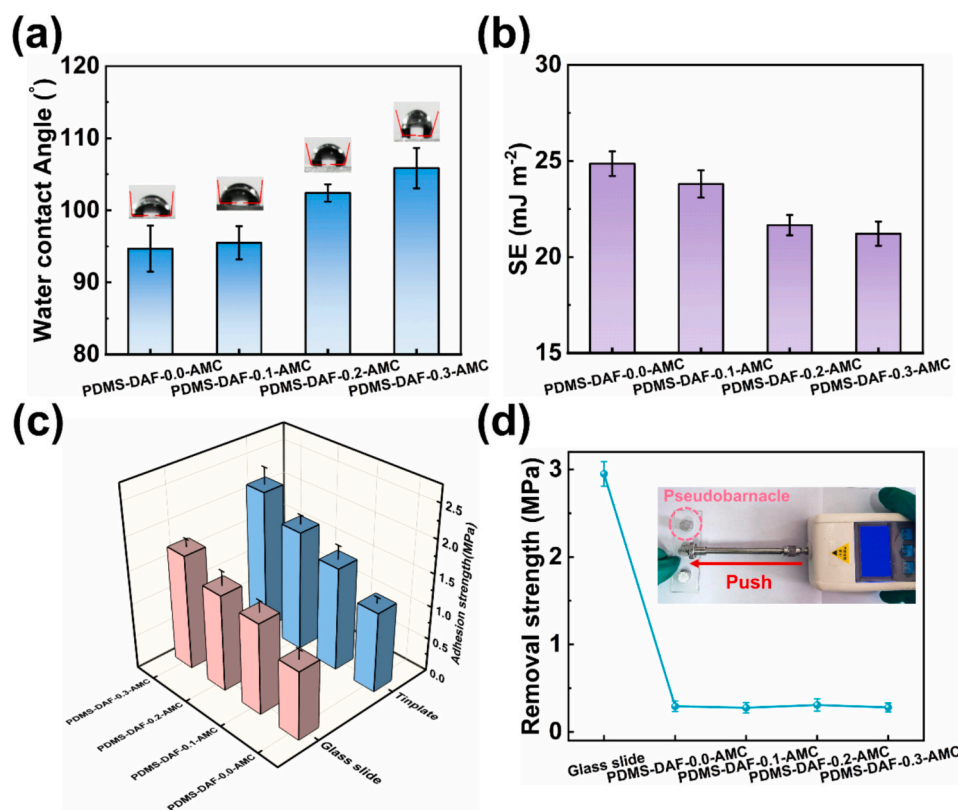


Fig. 5. (a) Water contact angle, and (b) SE of PDMS-DAF-x-AMC coatings. (c) Adhesion strength of PDMS-DAF-x-AMC coatings on tinplate and glass slide substrate. (d) Diagram of pseudobarnacle removal experiment and removal strength on glass slide and PDMS-DAF-x-AMC coatings.

increased surface roughness of the tinplate, achieved via sandpaper polishing. This process facilitates mechanical interlocking with the coating, thereby enhancing the adhesion strength.

The fouling release performance of coatings was further evaluated using pseudobarnacle removal tests, in which a lower removal strength signifies enhanced fouling release capability. As shown in Fig. 5(d), the removal strength of aluminum studs from the glass slide surface is notably high at 2.9 MPa. Conversely, the removal strength from the PDMS-DAF-x-AMC coating is considerably lower, ranging from 0.27 MPa to 0.3 MPa. This reduction is attributed to the coating's low surface energy properties, indicating its superior fouling release capability.

To assess the resistance to adhesion of abiotic pollutants, a self-cleaning test was conducted. Carbon powder served as the hydrophobic pollutant, while copper chloride dihydrate ($\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$) was employed as the hydrophilic pollutant. As illustrated in Fig. S8, positioning the slide at an inclination of 10° required only a minimal volume of water to effectively eliminate carbon powder contaminants and copper chloride particles from the coating surface. These findings indicate that the coating exhibits excellent self-cleaning properties, which are of considerable importance for the resistance of inorganic pollutants.

3.4. Antifouling properties

Fig. 6(a) and (c) depicts the antibacterial efficacy of the PDMS-DAF-x-AMC coating. The PDMS sample showed significant adhesion of *Pseudomonas* and *E. coli*. Importantly, the PDMS-DAF-0.0-AMC coating, which does not contain DAF, exhibited a marked reduction in bacterial colonies on its surface. This reduction is attributed to the bactericidal properties of the benzopyranone ring structure in AMC [51]. However, due to the low concentration of AMC, the bacteriostatic efficacy was limited to 78 % and 81 % against *Pseudomonas* and *E. coli*, respectively. With an increase in DAF content, the antibacterial efficacy of PDMS-DAF-0.1-AMC achieved a 90 % and 93 % reduction in bacterial

colonies for *Pseudomonas* and *E. coli*, respectively. Furthermore, the PDMS-DAF-0.2-AMC and PDMS-DAF-0.3-AMC coatings demonstrated bacteriostatic efficiencies of 99.6 % and 99.9 % against *Pseudomonas* and *E. coli*, respectively, with only a minimal presence of colonies. These findings indicate that PDMS-DAF-x-AMC demonstrates exceptional antibacterial properties, attributable to the synergistic interaction between the intrinsic antibacterial activity of DAF and the fluorescence response of AMC. To further verify the antibacterial mechanism, the morphology of *Pseudomonas* adhered to PDMS-DAF-0.0-AMC and PDMS-DAF-0.3-AMC was observed by SEM. As shown in Fig. S9, the bacteria attached to PDMS-DAF-0.0-AMC exhibited a relatively uniform morphology without significant collapse or deformation, indicating a viable state. In contrast, the bacteria attached to PDMS-DAF-0.3-AMC displayed pronounced morphological depressions and an overall shrunken, irregular appearance, consistent with a non-viable state. This further underscores the role of the DAF structure in inducing bacterial death by compromising the normal functions of the cell membrane.

During the initial phase of biofouling, algae play a crucial role in the development of macrofouling following the preliminary fouling stage. Diatoms (*N. incerta*) were chosen as representative marine algae to assess anti-diatom properties. We investigated the colonization and growth of *N. incerta* on the coating surface over a period of 7 d under natural light conditions with diurnal variations. This was compared to samples stored in the absence of light to simulate all-weather conditions, in order to evaluate the fluorescent antifouling efficacy of the coating. Fig. 6(b) and (d) clearly demonstrates that a substantial quantity of *N. incerta* adheres to the surface of PDMS, both under natural light conditions (511 ± 135 diatoms mm^{-2}) and when shielded from light (538 ± 187 diatoms mm^{-2}). This phenomenon can be attributed to the hydrophobic nature of PDMS, which possesses a water contact angle about 116.5° [28], aligning with the preference of diatoms for attachment to hydrophobic surfaces [52]. The preferential adhesion of diatoms to hydrophobic surfaces stems from their hydrophobic extracellular

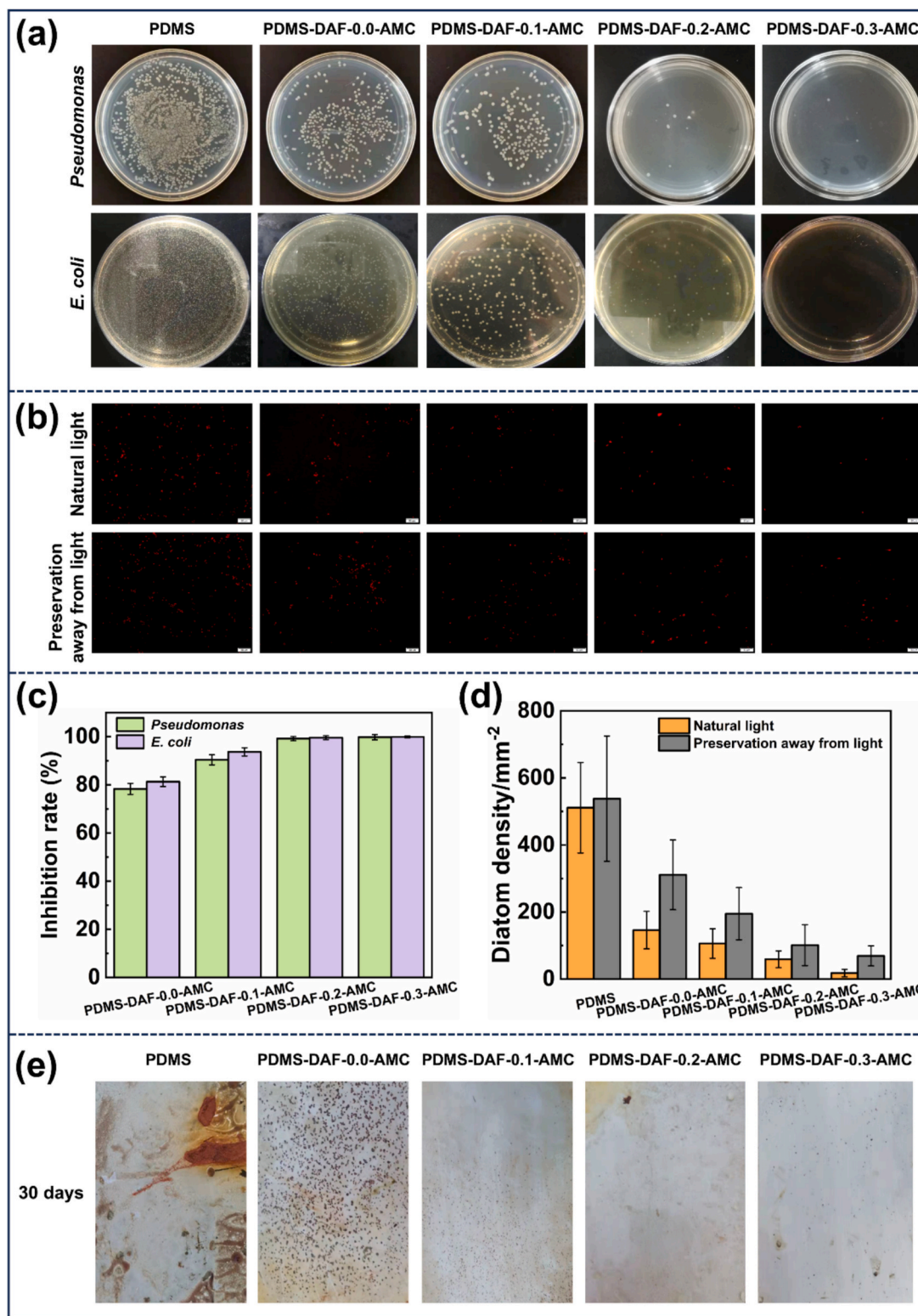


Fig. 6. (a) Optical images of colony formation. (b) Representative fluorescence images *N. incerta* adhered to each coating surface kept in natural light or preserved away from light after 7 days (the scale is 50 μm). (c) Antibacterial efficiency of PDMS-DAF-x-AMC. (d) Quantitative diatom settlement density of *N. incerta* on different coatings kept in natural light or preserved away from light after 7 d. (e) Images of different tested panels (150 mm $\times$ 250 mm $\times$ 3 mm) after immersion in seawater for 30 days.

polymeric substances (EPS), which exhibit stronger van der Waals interactions with low surface energy substrates like PDMS, as shown by Hoagland et al. [53]. Holland et al. [52] demonstrated that diatoms *Navicula* require significantly higher shear stress to detach from

hydrophobic surfaces compared to hydrophilic ones, highlighting surface energy compatibility. The adhesion complex optimizes EPS secretion for stable attachment on hydrophobic substrates, while hydrophobic surfaces also promote bacterial biofilms that enhance

diatom colonization [54]. The adhesion of diatoms to the PDMS-DAF-0.0-AMC surface was significantly reduced compared to the PDMS surface. This reduction can be attributed to the enhanced hydrophilic properties of PDMS-DAF-0.0-AMC, as evidenced by its water contact angle of 94.8° (Fig. 5(a)). Notably, even a minimal incorporation of AMC markedly enhanced the antifouling efficacy. Under sunlight, the PDMS-DAF-0.0-AMC surface demonstrated superior resistance to diatom adhesion, with a density of 146 ± 56 diatoms mm^{-2} . Similar behavior is observed in other PDMS-DAF-x-AMC coated surfaces when they are exposed to sunlight. The mechanism of fluorescent antifouling is predicated upon the interaction between cellular physiological activities and illumination, which affects the spatial interference between chloroplasts and holdfast-like structures within the thecae [55]. Therefore, under increased light conditions, diatoms prioritize photosynthesis over settlement [56]. In addition, diatom resistance increased with increase in DAF content. Although the water contact angle of the coating increases from $\sim 94.8^\circ$ for PDMS-DAF-0.0-AMC to 105.8° for PDMS-DAF-0.3-AMC, the diatom density on PDMS-DAF-0.3-AMC was measured to be 18 ± 11 diatoms mm^{-2} under natural light conditions and 69 ± 30 diatoms mm^{-2} when shielded from light. This substantial reduction can be ascribed to the biological activity of DAF on the coating surface. The inhibitory or bactericidal effect of the 1,2,5-oxadiazole structure is realized through its interaction with the cell wall and cell membrane systems [57,58]. These findings further emphasize fluorescent antifouling made a main contribution to the synergistic antifouling effect. The synergistic antifouling mechanism between DAF and AMC is rooted in their complementary biological targets and surface energy optimization. DAF's intrinsic antibacterial activity ($>99\%$ bacteriostatic efficiency, Fig. 6a and c) disrupts bacterial adhesion by targeting cell wall/membrane systems, while AMC's fluorescence under natural light selectively suppresses diatom settlement (18 ± 11 vs. 69 ± 30 diatoms mm^{-2} in darkness, Fig. 6b and d) by interfering with photosynthesis. This dual action operates sequentially: DAF inhibits biofilm formation, and AMC further destabilizes diatom adhesion even under static conditions. The balanced surface energy ($21.11\text{--}24.86$ mJ m^{-2} , Fig. 5b) and hydrophobicity (water contact angle: $94.8^\circ\text{--}105.8^\circ$, Fig. 5a) further reduce compatibility with both bacterial and diatom extracellular polymeric substances. While no direct molecular interaction occurs between DAF and AMC, their combined physicochemical and biological effects mimic coral's natural strategies, where chemical defense and light-mediated antifouling jointly suppress biofouling across multiple stages.

Surface roughness plays a key role in inhibiting the attachment of fouling organisms, as these organisms typically experience challenges in adhering to smooth surfaces. Furthermore, smooth surfaces contribute to improved fouling release performance [59]. Atomic Force Microscopy (AFM) was utilized to quantitatively assess the surface roughness of each coating (Fig. S10). PDMS exhibited an exceptionally low R_q of ~ 1.24 nm, indicating a notably smooth surface. Moreover, the R_q of PDMS-DAF-0.0-AMC, PDMS-DAF-0.1-AMC, PDMS-DAF-0.2-AMC, and PDMS-DAF-0.3-AMC rose to approximately ~ 11.2 , ~ 12.9 , ~ 15.6 , and ~ 20.7 nm, respectively. This phenomenon can be attributed to micro-phase separation resulting from the thermodynamic incompatibility between the polydimethylsiloxane (PDMS) soft segment and the hard segments of polyurethane and polyurea. It is noteworthy that as the DAF content increases, the extent of micro-phase separation on the coating surface progressively intensifies. This is attributed to the incremental formation of hard micro-regions induced by the interaction between the chain extender DAF and isocyanate, resulting in enhanced surface roughness. This phenomenon is also distinctly observable in the phase diagram. While an increase in the surface roughness of the coating is theoretically expected to enhance the adhesion of microorganisms, empirical evidence from prior antibacterial and anti-diatom experiments indicates otherwise. Notably, the PDMS-DAF-0.3-AMC coating maintains exceptional antifouling performance due to the synergistic effects of its antibacterial groups and fluorescence response.

Consequently, within our system, the impact of increased surface roughness on the reduction of antifouling properties is minimal, rendering its negative effects almost negligible.

A marine field test was conducted in Qingdao Port, China ($24^\circ 33'N$, $118^\circ 09'E$) to further confirm the static antifouling performance of the prepared coatings, the panel coated with PDMS elastomer was used as reference sample. Images of tested panels after immersion in seawater for 30 days are presented in Fig. 6e. After 30 days, it can be seen clearly that the panel coated with PDMS elastomer was covered with a large number of fouling organisms such as macroalgae, sponges and hydroids, indicating severe fouling pressure at the site. However, a large number of micro-fouling organisms were observed on the surface of PDMS-DAF-0.0-AMC samples, with no formation of macro-fouling organisms, emphasizing the importance of AMC's fluorescent response in inhibiting macro-fouling formation. As the content of DAF antibacterial groups increased, only a small amount of micro-fouling attachment was observed on PDMS-DAF-0.1-AMC samples, while PDMS-DAF-0.2-AMC and PDMS-DAF-0.3-AMC showed almost no micro-fouling attachment. This further demonstrates the synergistic anti-fouling effects of DAF antibacterial groups and AMC's fluorescent response. In summary, PDMS-DAF-0.2-AMC and PDMS-DAF-0.3-AMC samples exhibited excellent antifouling performance in real-sea immersion tests.

4. Conclusions

In conclusion, we have developed a coral-biomimetic PDMS marine antifouling coating (PDMS-DAF-x-AMC) that leverages intrinsic antibacterial properties and a fluorescent response to achieve synergistic antifouling effects. The incorporation of supramolecular interactions within the PDMS matrix enhances its mechanical properties, adhesion strength, and self-healing capabilities, thereby ensuring its durability in complex marine environments. Due to the low surface energy of PDMS, the coating demonstrates excellent fouling-release performance and self-cleaning properties. Additionally, the coating exhibits remarkable fouling resistance, attributed to the synergistic effects of the antibacterial activity of DAF and the fluorescence response of AMC. This study introduces a novel design concept for the development of high-performance antifouling coatings by integrating diverse environmentally sustainable antifouling strategies.

Future research will concentrate on assessing the long-term durability of the coating, incorporating extended field trials (lasting 12 months or more) to evaluate mechanical integrity, antifouling effectiveness under seasonal variations, resistance to UV degradation, and susceptibility to abrasive wear. The investigation will include an analysis of the stability of supramolecular interactions under cyclic stress, monitoring of zinc ion leaching, and quantification of fluorescence persistence during UV aging. The objective of these studies is to optimize the coating's resistance to weathering while preserving its environmentally friendly characteristics.

CRedit authorship contribution statement

Jiawen Sun: Writing – review & editing, Writing – original draft, Visualization, Supervision, Investigation, Formal analysis, Data curation, Conceptualization. **Susu Zhao:** Software, Investigation, Formal analysis, Data curation, Conceptualization. **Chengxin Song:** Writing – original draft, Data curation, Conceptualization. **Yuqing Zhu:** Visualization, Investigation, Data curation. **Jizhou Duan:** Supervision, Project administration, Funding acquisition. **Xiaofan Zhai:** Visualization, Supervision, Funding acquisition. **Baorong Hou:** Supervision, Project administration, Funding acquisition. **Chuanhui Gao:** Supervision, Resources. **Yuetao Liu:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2025.109357>.

Data availability

Data will be made available on request.

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